

AI Calling Agent Cost Reduction Plan

Version 2 — rebuilt on **verified provider rates** and **measured live calls**

Prepared for: Engineering & Founders

Status: Phase 1 shipped and verified. Phase 2 pending deployment.

Exchange rate: \$1 = ₹96 (verified 27 Jul 2026)

Headline. Cost per 3-minute call has been cut from **₹12.86 to ₹8.40 (~35%)**, already deployed and verified on live calls. Gross margin on the Starter plan rises from **59% to 68%**, and break-even drops from **128 to ~108 customers**. The remaining work is not a cost problem — it is a **hosting** problem.

What changed since v1. Three numbers in v1 were wrong and are corrected here: TTS was priced at \$0.05/1k characters (actual: **\$0.06**), the LLM was costed at ₹1.50/call (actual today: **₹0** on Gemini's free tier), and a TTS model switch was claimed to halve cost (it saves **nothing** — all Turbo models bill at one rate).

1. Verified rates — the basis for everything

Every figure below was confirmed against a provider price sheet or measured from our own call logs. Anything still estimated is flagged in **amber**.

Input	Rate	Status
MiniMax TTS, Turbo tier	\$60 / 1M chars = ₹0.00576 / character	verified — MiniMax pricing page
Deepgram STT (nova-2)	~\$0.0043 / min	verified
Gemini <code>gemini-flash-latest</code>	₹0 (free tier)	verified — currently in use
LiveKit Cloud agent session	~\$0.01 / min	verified
MiniMax voice cloning	\$1.5 / voice (one-time)	verified
WhatsApp utility (India)	~₹0.12 / msg; free in 24h window	verified
Groq <code>llama-3.3-70b</code> (alternative)	\$0.59 in / \$0.79 out per M tokens	verified
Vobiz telephony	₹0.30–0.50/min · DID ₹500–800/mo	ESTIMATED — no public pricing
Calls per tenant per month	300	ASSUMPTION

Single biggest open risk. Vobiz publishes no rates, so telephony (₹1.05/call plus ₹500/month per number) is the one unverified input in this model. Confirm the real per-minute and DID price before these numbers are used for pricing decisions.

Measured from a real call (30 Jul, 14:57–14:59)

Observation	Value	Meaning
Call duration	1.63 min	4 turns, booking completed
Agent speech, billable	610 chars	= ₹3.51 of TTS
Greeting (75 chars)	₹0	served from disk cache, <code>ttfb=0.003s</code>
Longest single reply	~250 chars	the waste the brevity rule removed

2. Cost per call — before vs after

Before optimisation

₹12.86

per 3-min call

Now (shipped)

₹8.40

−35%

If LLM becomes paid

₹9.90

plan for this

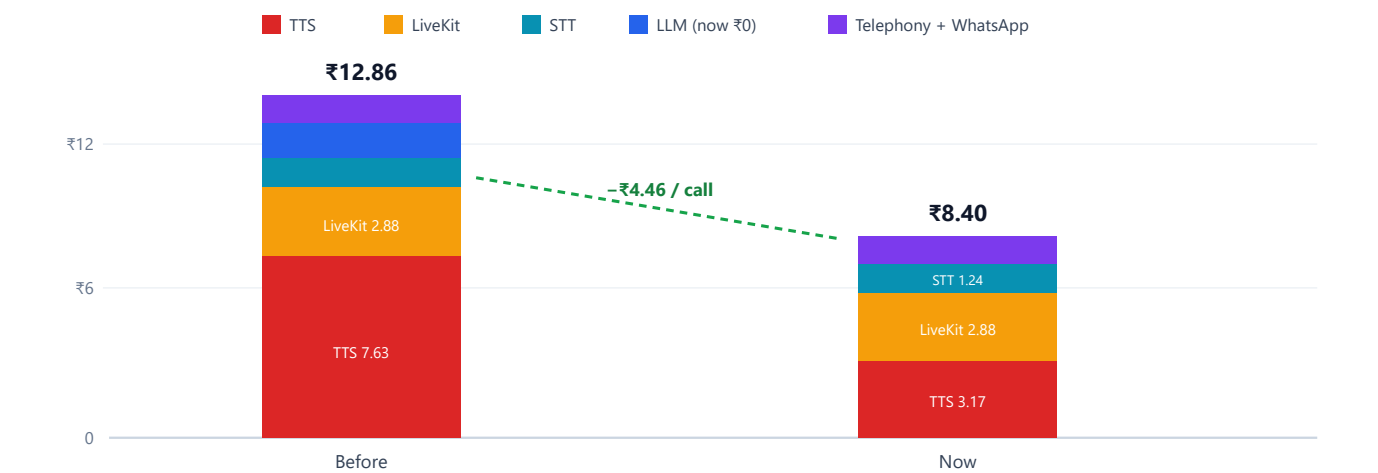


Figure 1 — TTS was 59% of the cost and is where the entire saving came from.

Component	Before	Now	How
TTS (MiniMax)	₹7.63	₹3.17	Greeting cached (₹0) + replies cut 250→110 chars
LiveKit Cloud	₹2.88	₹2.88	unchanged (see Phase 2)
STT (Deepgram)	₹1.24	₹1.24	unchanged
LLM	₹1.50	₹0	Gemini free tier (was mispriced in v1)
Telephony (est.)	₹1.05	₹1.05	needs a volume rate
WhatsApp	₹0.06	₹0.06	free inside the 24h window
Total / 3-min call	₹12.86	₹8.40	−35%
Per minute	₹4.29	₹2.80	

Formula: `TTS = characters spoken × ₹0.00576` . A cache hit costs ₹0.

3. What was actually changed (all verified on live calls)

#	Change	File	Saving	Evidence
1	Disk-backed TTS cache for fixed lines	<code>minimax_tts.py</code>	₹0.44	TTS cache hit (disk) , ttfb 721ms → 3ms
2	Brevity rule — one sentence, ≤25 words, answer only what was asked	<code>main.py</code>	₹2.25	reply 250 → 110 chars on a live call
3	KB rule — never recite a whole list/schedule	<code>main.py</code>	incl.	was reading both doctors' full week
4	System prompt trim — removed duplicated rules	<code>main.py</code>	quota	1,205 → 1,109 tokens; all 18 rules verified intact
5	TTS at 8 kHz (PSTN is 8 kHz anyway)	<code>main.py</code>	latency	ttfb 826ms → 573ms, 3× less data
6	LLM = ₹0 via Gemini free tier	<code>.env</code>	₹1.50	correction to v1's pricing

Reliability fixes found while measuring (not cost, but they were breaking calls)

Bug	Symptom on a real call	Fix
Tool call spoken as text	caller heard <code><function=end_call>{}/</function></code>	generalised the TTS strip filter — 8/8 tests pass
Abbreviation mispronounced	model wrote <code>डॉ.</code> , the voice said "दो" (= "two")	banned abbreviations + ASCII digits in the prompt
Bad cloned voice id	MiniMax returns HTTP 200 with no audio → whole call silent	auto-fallback to a known-good voice (tested live)
Preemptive generation	2 concurrent LLM requests → <code>Request timed out</code> → 13.4s replies	<code>AGENT_PREEMPTIVE=0</code>
Duplicate agent processes	LiveKit sent calls to a stale worker with old config	<code>agent/agent-procs.ps1</code> cleanup script
Stale dispatch after an outage	full session (TTS+STT+LLM) run against an empty room	guard: no participant ⇒ end job immediately

4. What is left is a hosting problem, not a code problem

After the changes above, every remaining complaint traced to **network distance and jitter** from running the agent on a home connection. Measured TCP-443 handshake times from the current machine:

Service	Min	Avg	Max	Effect on calls
Gemini	45ms	287ms	1,064ms	10s timeouts ⇒ 13.4s replies
Deepgram	301ms	378ms	675ms	STT socket dropped mid-call (1006)
MiniMax	68ms	138ms	433ms	0.6s gap between sentences = audible cut
LiveKit	75ms	139ms	291ms	8.5-minute total outage observed

The 8.5-minute outage matters commercially. While the worker was disconnected, LiveKit had nowhere to send calls — callers heard ringing and hung up. Then the queued dispatches arrived at once and two sessions ran in the same process, which is what "the audio is breaking up" actually was.

Fix: one Linux VM in an India region

Problem	Today (home)	On the VM
Gemini round trip	45–1,064ms (23× jitter)	~10–30ms, stable
Reply latency	2.4–13.4s	target < 1.5s
Audio gaps	every reply	expected to disappear
Context fetch per call	~1.9s	~50ms (Supabase is in Mumbai)
Windows <code>RtcError</code> panic	worker recycled every call	gone ⇒ warm worker
Two clients at once	untested / breaks	works (<code>AGENT_RECYCLE_AFTER_CALL=0</code>)
ngrok as webhook URL	unstable, a launch blocker	real domain + HTTPS

Constraints: **x86_64** (the LiveKit noise-cancellation library has no arm64 build), 2 vCPU minimum (real-time audio), Ubuntu 24.04, Mumbai/Bangalore. Full runbook in `docs/DEPLOYMENT.md` .

5. Monthly operating cost

Fixed (everything, including deployment)

Item	Monthly
VM — 2 vCPU / 8 GB, Mumbai (agent + backend + Redis)	₹610
Supabase Pro (Free tier auto-pauses — unusable in production)	₹2,400
Domain (amortised)	₹85
Frontend — Cloudflare Pages	₹0
TLS — Caddy / Let's Encrypt	₹0
Redis, logs — on the VM / journald	₹0
Total fixed	₹3,095

Total at scale

Total = ₹3,095 + tenants × (300 calls × ₹8.40 + ₹500 DID) → per tenant = ₹3,020/month

Tenants	Variable	Fixed	Total / month	Was (v1)
1 (testing)	₹3,020	₹3,095	₹6,115	—
10	₹30,200	₹3,095	₹33,295	—
100	₹3.02 L	₹3,095	₹3.05 L	₹4.19 L
500	₹15.1 L	~₹8,000	₹15.2 L	₹20.8 L
1,000	₹30.2 L	~₹15,000	₹30.4 L	₹41.5 L

The server is not the cost. At 100 tenants the ₹610 VM works out to **₹0.02 per call**. Optimising hosting further is wasted effort — the levers that matter are **characters spoken** (done) and a **Vobiz volume rate** (open).

6. Pricing and margin

Margins below assume a customer uses **65%** of their included calls, at the verified COGS of ₹8.40/call plus ₹500/month for the phone number.

Plan	Monthly	Yearly	Calls	COGS	Margin now	Was	Target customer
Starter	₹4,999	₹49,990	200	₹1,592	68%	59%	Single clinic, salon, spa
Professional	₹12,999	₹1,29,990	750	₹4,599	65%	51%	Multi-doctor clinics, real estate
Business	₹29,999	₹2,99,990	2,500	₹14,150	53%	41%	Multi-location chains, hospitals
Enterprise	₹79,999+	custom	8,000	₹44,680	44%	39%	BPOs, aggregators, white-label

Usage-based option	Price	COGS	Margin
Pay-as-you-go, per call	₹20	₹8.40	58%
Per minute	₹7	₹2.80	60%
Overage beyond plan	₹12–25	₹8.40	30–66%
Custom voice setup (one-time)	₹2,000–5,000	₹144 + labour	high

Break-even

Fixed opex ₹6,00,000 ÷ (ARPU ₹9,000 × 62% margin) = **~108 customers** — down from 128 in v1, purely from the cost work.

Tenants	Revenue	COGS	Opex	Net / month
100	₹9.0 L	₹3.05 L	₹6.0 L	–₹0.05 L
500	₹45 L	₹15.2 L	₹12 L	+₹17.8 L
1,000	₹90 L	₹30.4 L	₹20 L	+₹39.6 L

7. Risks and open items

Risk	Impact	Action
Vobiz rates unknown	₹1.05/call + ₹500/DID is the only unverified input in this model	Get a written per-minute + DID rate; negotiate volume
Gemini free tier ends	+₹1.50/call → ₹9.90; Starter margin 68% → 64%	Budget for it; Groq is cheaper but skipped the booking intake
MiniMax 2.6-turbo is "Legacy"	forced migration to 2.8 eventually	Re-run the listening test when it happens; price is identical
Leaked secrets in git history	JWT secret, WhatsApp token, old DB password	Rotate before deploying (checklist blocker #1)
No Supabase RLS	tenant isolation is app-code only	Add RLS before onboarding real clients
Quota not enforced	<code>ENFORCE_CALL_QUOTA=false</code> ⇒ margin leaks	Turn on, fail-closed with a grace buffer
FX exposure	costs in USD, revenue in INR; rupee fell ~10% in a year	Price with headroom

8. Next steps, in order

#	Action	Effort	Payoff
1	Rotate the leaked secrets	30 min	removes a launch blocker
2	Deploy to a Mumbai VM (<code>docs/DEPLOYMENT.md</code>)	1 day	fixes latency, audio gaps, outages, concurrency
3	Get real Vobiz rates	a call	removes the last unknown from this model
4	Set the plan prices in <code>plans.py</code> (currently <code>null</code>)	1 hour	unblocks Razorpay checkout
5	Enable call-quota enforcement	2 hours	stops margin leakage
6	Investigate MiniMax audio subscription (20% off)	a call	TTS is still the largest line item

Summary for the team. Cost per call is down 35% (₹12.86 → ₹8.40) with no loss of quality, and six call-breaking bugs were found and fixed along the way. Margins are now 53–68% across plans and break-even is ~108 customers. The next win is not in the code — it is moving off the home connection onto a ₹610/month Mumbai VM.

Method. Rates verified against provider pricing pages (MiniMax, Deepgram, LiveKit, Groq, Supabase, Google) on 27–30 Jul 2026; FX \$1 = ₹96. Per-call figures derived from our own agent logs (character counts, `TTSMetrics` , `LLMMetrics` , `EOUMetrics`) on live calls to `+918065480571` .

Assumptions: 3-minute average call, ~6 turns, 300 answered calls per tenant per month, 65% allowance utilisation. **Estimated (not verified):** Vobiz per-minute and DID pricing, taken from Indian market comparables. Planning estimates for internal discussion, not a billing guarantee.